

Practical Assignment: Exploring AWS → Snowflake Data Ingestion Use Cases

(Questions Only – Hands-On Exploration)

❑ Use Case 1: One-Time Manual Data Load (Access Key Method)

Scenario:

A team needs to load historical CSV files from S3 into Snowflake **one time only**.

How It Works

S3 Bucket → Stage (with AWS Keys) → Snowflake Table

Step 1: You have files in S3

Step 2: Create External Stage in Snowflake

- Provide AWS_ACCESS_KEY_ID
- Provide AWS_SECRET_ACCESS_KEY

Step 3: Snowflake uses these keys to authenticate with AWS

Step 4: Data loads from S3 → Snowflake Table

Exploration Questions

1. What Snowflake objects are mandatory to perform a one-time manual load?

TABLE → Where data goes |

FILE FORMAT → How to read file |

STAGE → S3 connection with keys

1.2 create table:

create or replace table employeee(emp_id int primary key,Name varchar (20),department varchar (20),Salary int,manager_name varchar(20),location varchar(20),UAN_NO INT);

Results (just now)	
Table	Chart
1 row	180ms
1	Table EMPLOYEEE successfully created.

1.3--External Stage Create (Access Key Method)

CREATE OR REPLACE STAGE et_stage

URL = 's3://decbucket1209/S3_file/employee_data_30_records.csv'

CREDENTIALS = (

AWS_KEY_ID = 'AKIAQHDWMTVPAA5XGQW7'

AWS_SECRET_KEY = 'H3q0t7+9RRBM8MA1VuGRS79Rx/3J+pkzwVolCgTB'

)

FILE_FORMAT = (

TYPE = 'CSV',

FIELD_DELIMITER = ',',

SKIP_HEADER = 1

);

Results (just now)	
Table	Chart
1 row	568ms
#	status
1	Stage area ET_STAGE successfully created.

1.4 --Validate Stage (LIST Command)

LIST @et_stage;

Results (just now)				
Table	Chart	1 row	506ms	
#	name	size	md5	last_modified
1	s3://decbucket1209/S3_file/employee_data_30_records.csv	1622	f11eecacfd24007bf283075593138483	Sun, 28 Dec 2025 08:43:10 GMT

1.5 --Load Data using COPY INTO

copy into employeee from @et_stage

FILE_FORMAT=(TYPE='CSV',SKIP_HEADER=1);

Results (just now)							
Table	Chart	1 row	3.3s				
#	file	status	# rows_parsed	# rows_loaded	# error_limit	# errors_seen	first_error
1	s3://decbucket1209/S3_file/employee_data_30_records.csv	LOADED	30	30	1	0	null

1.6 validate

select*from employeee

Results (just now)							
Table	Chart	30 rows	142ms				
#	EMP_ID	NAME	DEPARTMENT	SALARY	MANAGER_NAME	LOCATION	UAN_NO
1	1	Employee_1	Finance	32500	Anand	Bangalore	1002003001
2	2	Employee_2	Finance	35000	Ram	Hyderabad	1002003002
3	3	Employee_3	Sales	37500	Suresh	Pune	1002003003
4	4	Employee_4	Operations	40000	Anand	Mumbai	1002003004
5	5	Employee_5	IT	42500	Ramesh	Chennai	1002003005
6	6	Employee_6	HR	45000	Kishore	Bangalore	1002003006
7	7	Employee_7	Finance	47500	Ram	Hyderabad	1002003007

2.Where and how are AWS credentials stored in Snowflake for this method?

```
CREATE STAGE my_stage

URL = 's3://bucket/'

CREDENTIALS = (

  AWS_KEY_ID = 'AKIAXXXXXXXX'    -- ← Access Key

  AWS_SECRET_KEY = 'xxxxxxxxx'    -- ← Secret Key

);
```

Results (just now)

Table Chart 1 row 89ms

status
Stage area ET_STAGE successfully created.

DESC STAGE et_stage;

-- or

SHOW STAGES;

Results (1 minute ago)

Table Chart 3 rows 30ms

created_on	name	database_name	schema_name	url	has_credentials	has_encryption_key
5-12-28 06:49:09.985 -0800	ET_STAGE	TAB_DB	AWS_SCHEMA	s3://decbucket1209/S3_file/employee_data_30_records.c	Y	N
5-12-28 01:14:51.229 -0800	EXT_STAGE2	TAB_DB	AWS_SCHEMA	s3://decbucket1209/S3_file/	N	N
5-12-28 02:59:38.216 -0800	MY_STG	TAB_DB	AWS_SCHEMA	s3://snowpipecon1209/pipefile/	N	N

3.What minimum S3 permissions are required for successful loading?

list @et_stage;

(1 minute ago) ... x

Table Chart 1 row 1.0s

name	size	md5	last_modified
s3://decbucket1209/S3_file/employee_data_30_records.	1622	f11eecacfd24007bf283075593138483	Sun, 28 Dec 2025 08:43:10 GMT

Permission	Purpose
s3: GetObject	Read files
s3: GetObjectVersion	Read versioned files
s3: ListBucket	List files

hari_user
Info
Delete

Summary

ARN
arn:aws:iam::015280676190:user/hari_user

Console access
Disabled

Access key 1
AKIAQHDWMTVPA5XGQW7 - Active
Used 10 hours ago. Created yesterday.

Created
December 28, 2025, 19:26 (UTC+05:30)

Last console sign-in
-

Access key 2
Create access key

Permissions
Groups
Tags
Security credentials
Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.
Remove
Add permissions

Search
Filter by Type
All types

Policy name	Type	Attached via
AmazonS3ReadOnlyAccess	AWS managed	Directly

4.What happens if credentials are revoked after the stage is created?

If I delete AWS keys, does stage work?"

Answer: Stage FAILS

Before Revoke: LIST @my_stage; → Works

After Revoke: LIST @my_stage; → Access Denied

Before vs After Revocation

BEFORE Revoke (Working)

List @et_stage

Results (just now)

Table
Chart
1 row 539ms

id	name	size	md5	last_modified
1	s3://decbucket1209/S3_file/employee_data	1622	f11eecacfd24007bf283075593138483	Sun, 28 Dec 2025 08:43:10 GMT

AFTER Revoke (Broken)

Access keys (1)

[Create access key](#)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

AKIAQHDWMTVPAASXGQW7	Actions ▼
Description -	Status ⊖ Inactive
Last used 11 minutes ago	Created Yesterday
Last used region ap-southeast-2	Last used service s3

list @et_stage;

0 rows ⓘ 970ms
 Failure using stage area. Cause: [The AWS Access Key Id you provided is not valid.]

5. How can you confirm which files were successfully loaded?

Answer: Use **COPY_HISTORY** function

```
SELECT file_name, status, row_count, first_error_message
FROM TABLE(INFORMATION_SCHEMA.COPY_HISTORY(
  TABLE_NAME => 'employeee',
  START_TIME => DATEADD(hour, -48, CURRENT_TIMESTAMP())
));
```

Results (just now) ⌵				
Table	Chart	1 row ⓘ 253ms ⬇ ⌚		
id	FILE_NAME	STATUS	ROW_COUNT	FIRST_ERROR_MESSAGE
1		Loaded	30	null

6. What happens if the same file is re-uploaded to S3 and reloaded?

Answer: File is Skipped by default

1st COPY: sales.csv →  LOADED (1000 rows)

2nd COPY: sales.csv →  SKIPPED (already loaded)

copy into employee from @et_stage
FILE_FORMAT=(TYPE='CSV',SKIP_HEADER=1);

Results 1 of 1 Row • Cluster 1

Export Results

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_
1	s3://decbucket1209/S3_file/employee_data_30_records.csv	LOADED	30	30	1	0	null	

copy into employee from @et_stage
FILE_FORMAT=(TYPE='CSV',SKIP_HEADER=1);

Results (1 minute ago)			
Table		Chart	
		1 row 744ms	
A status			
1	Copy executed with 0 files processed.		

- 7. How does Snowflake prevent or allow duplicate loads?
- 8. What security risks are visible during implementation?

Risk	Problem
1	Keys visible in SQL
2	Keys in QUERY_HISTORY
3	No auto key rotation
4	Keys don't expire

SELECT query_text FROM SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY
WHERE query_text ILIKE '%AWS_SECRET_KEY%';

Results (1 minute ago)			
Table		Chart	
		11 rows 3.9s	
A QUERY_TEXT			
1	create or replace stage et_stage url='s3://decbucket1209/S3_file/employee_data_30_records.csv' credentials =(aws_key_id= '...',aws_secret_key='...');		
2	create or replace stage et_stage url='s3://decbucket1209/S3_file/employee_data_30_records.csv' credentials =(aws_key_id= '...',aws_secret_key='...');		
3	create or replace stage et_stage url='s3://decbucket1209/S3_file/employee_data_30_records.csv' credentials =(aws_key_id= '...',aws_secret_key='...' FILE_FORMAT =(		
4	create or replace stage et_stage url='s3://decbucket1209/S3_file/employee_data_30_records.csv' credentials =(aws_key_id= '...',aws_secret_key='...')		
5	create or replace stage et_stage url='s3://decbucket1209/S3_file/employee_data_30_records.csv' credentials =(aws_key_id= '...',aws_secret_key='...') create or replace :		
6	CREATE OR REPLACE STAGE et_stage URL = 's3://decbucket1209/S3_file/employee_data_30_records.csv' CREDENTIALS = (AWS_KEY_ID = '...' AWS_SECRET_KEY		

- 9. How would this approach behave in a production environment?
answer: **NOT Recommended**

Problem	Impact
Key rotation	Must manually update ALL stages
Security audit	✗ Fails compliance requirements
Key compromise	Hard to track & revoke all usages
Team scaling	Keys get shared dangerously
No expiration	Breach = unlimited access window

OK TO USE:

- One-time data loads
- POC / Proof of Concept
- Development / Testing only

NEVER USE:

- Production workloads
- Recurring data pipelines
- Sensitive/PII data
- Compliance-regulated environments

Production Alternative: IAM Roles

-- Use this instead for production

CREATE STAGE my_stage

URL = 's3://bucket/'

STORAGE_INTEGRATION = my_s3_integration; -- ☒ No keys!

10. Why is this method discouraged for long-term use?

**question meaning Why use Storage Integration instead?:
Storage Integration is More Secure**

ACCESS KEYS vs STORAGE INTEGRATION	
❌ ACCESS KEYS	✅ STORAGE INTEGRATION
• Keys in Snowflake • Manual rotation • Visible in logs • Per-stage credentials • Hard to audit • Security risk	• No keys in Snowflake • AWS handles rotation • IAM Role based • One integration, many stages • CloudTrail logging • Secure & compliant
RECOMMENDED APPROACH: <pre> -- Create Storage Integration (one-time) CREATE STORAGE INTEGRATION s3_int TYPE = EXTERNAL_STAGE STORAGE_PROVIDER = 'S3' STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::123:role/sf-role' STORAGE_ALLOWED_LOCATIONS = ('s3://bucket/'); -- Create Stage WITHOUT credentials CREATE STAGE secure_stage URL = 's3://bucket/' STORAGE_INTEGRATION = s3_int; </pre> ← No keys needed!	

❑ Use Case 2: Secure Batch Loading (Storage Integration + COPY INTO)

S3 — IAM Role — Storage Integration — Stage — Table

(No credentials stored in Snowflake!)

Scenario:

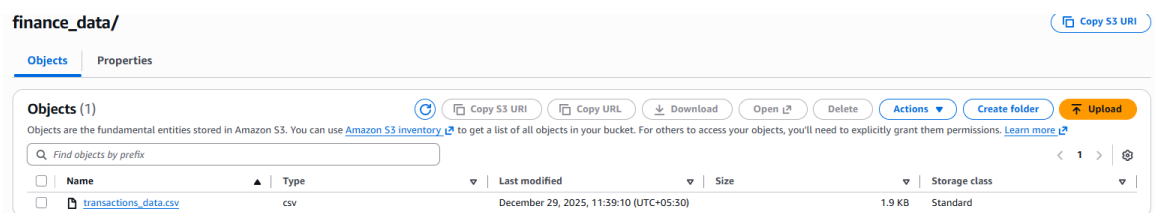
A finance team loads daily CSV files securely from S3 using IAM roles.

Exploration Questions

1. What AWS and Snowflake components are involved before loading begins?

AWS Components:

Amazon S3 Bucket – stores daily finance CSV files



IAM Role – grants Snowflake permission to read S3

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (2) [Info](#)

You can attach up to 10 managed policies.

Filter by Type

Search All types

☐	Policy name	Type	Attached entities
☐	AmazonS3ReadOnlyAccess	AWS managed	2
☐	AmazonS3TablesReadOnlyAccess	AWS managed	1

IAM Trust Policy – allows Snowflake to assume the role

Trusted entities [Edit trust policy](#)

Entities that can assume this role under specified conditions.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::974916068036:user/externalstages/ciicac0000"
8       },
9       "Action": "sts:AssumeRole",
10      "Condition": {}
11    }
12  ]
13 }
```

S3 Bucket Policy – restricts access to specific paths

Snowflake Components

Storage Integration – secure bridge between Snowflake and AWS

create or replace storage integration finance_int

type= EXTERNAL_STAGE

ENABLED= TRUE

STORAGE_PROVIDER ='S3'

STORAGE_AWS_ROLE_ARN ='arn:aws:iam::015280676190:role/fiance'

STORAGE_ALLOWED_LOCATIONS =('s3://financeteam129/finance_data/');

Results (just now)

Table	Chart
status	
1	Integration FINANCE_INT successfully created.

External Stage – points to the S3 location using the integration

Points to the S3 location using the storage integration.

Target Table – finance data table

Finance transactions table where data is loaded.

2. How does Snowflake assume the IAM role during loading?

How does Snowflake get AWS access without keys?

Trust Relationship + STS AssumeRole

Snowflake AWS Account — AssumeRole — Your IAM Role — S3

STEP 1: Get Snowflake's AWS Account & External ID :

DESC STORAGE INTEGRATION my_s3_integration;

	property	property_type	property_value	property_default
1	ENABLED	Boolean	true	false
2	STORAGE_PROVIDER	String	S3	
3	STORAGE_ALLOWED_LOCATIONS	List	s3://decbucket1209/S3_file/	[]
4	STORAGE_BLOCKED_LOCATIONS	List		[]
5	STORAGE_AWS_IAM_USER_ARN	String	arn:aws:iam::974916068036:user/external	
6	STORAGE_AWS_ROLE_ARN	String	arn:aws:iam::015280676190:role/R1	
7	STORAGE_AWS_EXTERNAL_ID	String	JK59297_SFCRole=2_TIXZWTh9hP6x6Ttb	
8	USE_PRIVATELINK_ENDPOINT	Boolean	false	false
9	COMMENT	String		

STEP 2: Configure Trust Policy in AWS IAM Role

Trusted entities

Entities that can assume this role under specified conditions.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::974916068036:user/externalstages/ciicac0000"
8       },
9       "Action": "sts:AssumeRole",
10      "Condition": {}
11    }
12  ]
13 }
```

STEP 3: Snowflake automatically assumes role during COPY

COPY INTO employee FROM @stage; → Snowflake calls AssumeRole

3. What is the purpose of STORAGE_ALLOWED_LOCATIONS?

Why do I specify allowed S3 paths?

Security boundary-limit access

create or replace storage integration my_s3_integration

type= EXTERNAL_STAGE

ENABLED= TRUE

STORAGE_PROVIDER ='S3'

STORAGE_AWS_ROLE_ARN

= 'arn:aws:iam::015280676190:role/R1'

STORAGE_ALLOWED_LOCATIONS

= ('s3://decbucket1209/S3_file/'); → only this path allowed

Results (just now)	
Table	Chart
1 row	98ms
id	status
1	Integration MY_S3_INTEGRATION successfully created.

Even if IAM role has broader S3 access,Snowflake restricts to ALLOWED_LOCATIONS only!

4.What happens if Snowflake tries to access a path outside the allowed location?

What if I try to access unauthorized S3 path?

Answer: Access Denied

Storage Integration allows: s3://bucket/finance/



You try to access: s3://bucket/hr-data/ ← Not allowed!



Result: **✗ ACCESS DENIED**

CREATE STAGE unauthorized_stagdde

URL = 's3://bucket/hr-data/'

STORAGE_INTEGRATION = finance_int;

Results (just now)	
0 rows	41ms



SQL compilation error: Location 's3://bucket/hr-data/' is not allowed by integration FINANCE_INT. Please use DESC INTEGRATION to check out allowed and blocked locations.

5. Can multiple stages share the same storage integration? Why?

Can I reuse one integration for many stages?

YES - Recommended Practice

6.How does Snowflake track loaded vs unloaded files?

How does Snowflake know which files are already loaded?

Load Metadata (64-Day Tracking)

```
SELECT file_name, status, row_count, last_load_time
FROM TABLE(INFORMATION_SCHEMA.COPY_HISTORY(
  TABLE_NAME => 'FINANCE_Transactions',
  START_TIME => DATEADD(days, -14, CURRENT_TIMESTAMP())
));
```

	FILE_NAME	STATUS	ROW_COUNT	LAST_LOAD_TIME
1		Loaded	20	2025-12-28 23:28:27.713 -0800

Metadata retained for 64 DAYS

7.What happens when new files are added to the same S3 folder?

If I add new files, will COPY load only new ones?

YES - Only NEW files are loaded

COPY INTO FINANCE_TRANSACTIONS FROM @EXT_STAGE3

FILE_FORMAT=(TYPE='CSV',SKIP_HEADER=1);

	status
1	Copy executed with 0 files processed.

8.How does this method compare to access-key loading in terms of security?

Why is Storage Integration more secure?

Much More Secure

SECURITY FLOW COMPARISON:

Access Keys:

SQL contains keys → Stored forever → Security risk

Storage Integration:

AssumeRole → Temp token (15min) → Auto-expires

ACCESS KEYS	STORAGE INTEGRATION
keys in Snowflake SQL	No keys stored anywhere
Keys in query history	IAM Role based (temporary)
Manual key rotation	AWS handles rotation

Keys never expire	Temporary STS tokens
Hard to audit	CloudTrail logging
Fails compliance	SOC2/HIPAA compliant

9.How can this method be automated using Snowflake Tasks?

How to schedule automatic daily loading?

Create Task with COPY INTO

10.What production-level advantages does this approach provide?

Why is this better for production?

6 Key Production Benefits

Advantage	Benefit
No Credentials	Nothing to rotate/manage
IAM Role Based	Follows AWS best practices
Centralized	One integration, many stages
Auditable	CloudTrail + Snowflake logs
Compliant	SOC2, HIPAA, PCI-DSS ready
Scalable	Easy to add new stages

COMPLETE PRODUCTION ARCHITECTURE:

S3 Bucket → IAM Role→ ☐ Storage Integration (ALLOWED_LOCATIONS) → ☐ External Stage →

Task (Scheduled) — ☐ COPY INTO — ☐ Table

Storage integration is secure,automated,auditable,scalable

□ Use Case 3: Near Real-Time Ingestion (Snowpipe)

Scenario:

An application generates CSV files every few minutes and requires **automatic ingestion** into Snowflake.

Exploration Questions

1. What triggers Snowpipe to start loading files automatically?

File Upload → S3 Bucket → S3 Event Notification → SQS Queue → Snowpipe (AUTO_INGEST = TRUE)

How it works:

→ Files are uploaded to S3 bucket

→ S3 sends event notification to SQS queue

→ Snowpipe automatically detects new files

→ Data loads automatically into Snowflake table

Step 1: Create Snowpipe with AUTO_INGEST

--USING SNOWPIPE

CREATE OR REPLACE PIPE PIPE1

AUTO_INGEST = TRUE AS

COPY INTO newly_emp from @my_stg

FILE_FORMAT = (TYPE = 'CSV', FIELD_OPTIONALLY_ENCLOSED_BY = ''',

FIELD_DELIMITER = ',', NULL_IF = ('NULL', 'null'),

EMPTY_FIELD_AS_NULL = TRUE, TRIM_SPACE = TRUE, SKIP_HEADER = 1

Results (just now)	
Table	Chart
1 row 3.9s	
status	
1	Pipe PIPE1 successfully created.

Step 2: Get the SQS Queue ARN

DESC PIPE PIPE1;

Event notifications (1)

Edit

Delete

Create event notification

Send a notification when specific events occur in your bucket. [Learn more](#)

☐	Name	Event types	Filters	Destination type	Destination
☐	PIPE	All object create events	-	SQS queue	arn:aws:sqs:ap-southeast-2:974916068036:sf-snowpipe-AIDAGF7MGULCL6KCTTRIV-SdviGsPJL4yH2r_bwY0Oag

Amazon EventBridge

Edit

For additional capabilities, use Amazon EventBridge to build event-driven applications at scale using S3 event notifications. [Learn more](#) or [see EventBridge pricing](#)

Send notifications to Amazon EventBridge for all events in this bucket

Off

3. Which Snowflake object listens to file arrival events?

How its worked:

Step 1: S3 receives new file

Step 2: S3 sends notification to SQS queue

Step 3: Pipe's notification_channel picks up SQS message

Step 4: Pipe automatically executes COPY command

Step 5: Data loads into target table

CREATE OR REPLACE PIPE PIPE1

```
AUTO_INGEST = TRUE
```

```
AS
```

```
COPY INTO newly_emp
```

```
FROM @my_stg
```

```
FILE_FORMAT = (
```

```
  TYPE = 'CSV',
```

```
  FIELD_OPTIONALLY_ENCLOSED_BY = '"',
```

```
  FIELD_DELIMITER = ',',
```

```
  NULL_IF = ('NULL', 'null'),
```

```
  EMPTY_FIELD_AS_NULL = TRUE,
```

```
  TRIM_SPACE = TRUE,
```

```
  SKIP_HEADER = 1
```

```
);
```

Results (just now)

TableChart

1 row3.1s

status

1Pipe PIPE1 successfully created.

The PIPE object listens to file arrival events.

Key components:

- **NOTIFICATION_CHANNEL:** The SQS queue ARN that receives S3 events
 - **AUTO_INGEST = TRUE:** Enables automatic listening
 - **COPY INTO definition:** Specifies what to do when event arrives
4. Does Snowpipe require a virtual warehouse? Why?
NO, Snowpipe does NOT require a virtual warehouse.

BATCH LOADING (COPY INTO)

Uses YOUR Virtual Warehouse

Stage (Files) → YOUR VIRTUAL WAREHOUSE → TABLE

(You manage & pay for warehouse compute time)

Key Points:

You control when the load happens (manual or scheduled)

Uses compute resources from your warehouse (X-Small, Small, Medium, etc.)

You pay for the warehouse runtime while loading data

Good for loading large volumes at once

Cost Model: You pay for warehouse credits consumed during the load operation

SNOWPIPE:

Uses Snowflake-Managed Serverless Resources:

Stage (Files) → SNOWFLAKE MANAGED COMPUTE → TABLE

(Serverless, automatic, no warehouse needed)

Key Points:

Automatically loads data as files arrive in the stage

No warehouse required - uses Snowflake's serverless compute

Event-driven (triggered by file notifications)

Continuous, near real-time loading

Snowflake manages all the compute resources

Cost Model: You pay per file size loaded (charged separately from warehouse credits)

Batch Loading = You control timing, you manage warehouse, you pay for warehouse time

Snowpipe = Automatic loading, serverless, pay per data loaded

Batch_loading:

COPY INTO students

```

FROM @security_stg//stuerr.csv

FILE_FORMAT = (

TYPE = 'CSV'

SKIP_HEADER = 1

)

VALIDATION_MODE = 'RETURN_ERRORS';

```

Results (just now)

	QUERY_TEXT	WAREHOUSE_NAME	EXECUTION_TIME	BYTES_SCANNED
1	COPY INTO students FROM @security_stg//stuerr.csv FILE_FORMAT = (TYPE = 'CSV' SKIP_HEAD	SMALL_WH	145	
2	copy into security_db.security.movies from @security_db.security.security_stg/movies_dataset.csv	COMPUTE_WH	306	
3	copy into security_db.security.movies from @security_db.security.security_stg/movies_dataset.csv	COMPUTE_WH	490	
4	copy into security_db.security.movies from @security_db.security.security_stg/movies_dataset.csv	COMPUTE_WH	0	
5	COPY INTO students FROM @security_stg/security.csv FILE_FORMAT = (TYPE = 'CSV' SKIP_HE	SECURITY_WH	279	

snowpipe:

```
--USING SNOWPIPE
```

```
CREATE OR REPLACE PIPE PIPE1
```

```
AUTO_INGEST = TRUE
```

```
AS
```

```
COPY INTO newly_emp
```

```
FROM @my_stg
```

```
FILE_FORMAT = (
```

```
    TYPE = 'CSV',
```

```
    FIELD_OPTIONALLY_ENCLOSED_BY = '',
```

```
    FIELD_DELIMITER = ',',
```

```
    NULL_IF = ('NULL', 'null'),
```

```
    EMPTY_FIELD_AS_NULL = TRUE,
```

```
    TRIM_SPACE = TRUE,
```

```
    SKIP_HEADER = 1
```

```
);
```

DESC PIPE PIPE1;

Results (just now)

	PIPE_NAME	FILE_NAME	STATUS	ROW_COUNT	FIRST_COMMIT_TIME
1	PIPE1	stuerr.csv	Load failed	0	null
2	PIPE1	employee_data_30_records.csv	Loaded	30	null
3	PIPE1	EMP.csv	Loaded	30	null
4	PIPE1	emppp.csv	Loaded	30	null

5. How does Snowpipe ensure files are loaded only once?

6. What happens if Snowpipe is paused?

BEFORE PAUSE:

Files → S3 → SQS → Snowpipe → Table ✓

DURING PAUSE:

Files → S3 → SQS → [STUCK!] → Table ✗



Messages accumulate in queue

AFTER RESUME:

Queued messages → Snowpipe → Table ✓

--Pause

ALTER PIPE pipe1 SET PIPE_EXECUTION_PAUSED = TRUE;

Results (just now)

	SYSTEM\$PIPE_STATUS('PIPE1')
1	{"executionState":"PAUSED","pendingFileCount":0,"notificationChannelName":"arn:aws:sqs:ap-southeast-2:974916068036:sf-snowpipe-AIDA6F7MGULCL6KCTTRIV-5dviGsPjI"}

--Resume:

ALTER PIPE pipe1 SET PIPE_EXECUTION_PAUSED = FALSE;

(just now)

	SYSTEM\$PIPE_STATUS('PIPE1')
1	{"executionState":"RUNNING","pendingFileCount":0,"notificationChannelName":"arn:aws:sqs:ap-southeast-2:974916068036:sf-snowpipe-AIDA6F7MGULCL6KCTTRIV-5dviGsPjI"}

checking pause and running commend:

SELECT SYSTEM\$PIPE_STATUS('pipe1');

When Snowpipe is paused:

Aspect	What Happens
New file events	Queue in SQS (not lost)
Processing	Stops completely
On resume	Queued files are processed
Risk	SQS messages expire after 4-14 days
Recovery	Use ALTER PIPE REFRESH if messages expire

7. How are failed files handled in Snowpipe?

What Happens When a File Fails?

File arrives → Snowpipe tries to load → ERROR

1. File marked as FAILED
2. Error saved in COPY_HISTORY
3. File stays in S3 (not deleted)
4. NO retry - you must fix it manually
5. Pipe continues with next file

Option	What It Does
ABORT_STATEMENT	One error → whole file skipped
CONTINUE	Skip bad rows, load good rows
SKIP_FILE_n	Skip file if errors > n
SKIP_FILE_n%	Skip file if error % > n

8. How can you monitor Snowpipe ingestion status?

9. What are the cost implications of Snowpipe vs batch loading?

10. In which scenarios is Snowpipe the best architectural choice?

❑ Use Case 4: Querying Data Lake Directly (External Tables)

Scenario:

A data analytics team wants to query large historical datasets **without moving data from S3**.

Exploration Questions

1. Where does the data physically reside when using external tables?
Is data copied to Snowflake or stays in S3?"

Data stays in s3

REGULAR TABLE (Internal Table)

Flow: S3 —❑ COPY INTO —❑ Snowflake Storage (Micro-partitions)

EXTERNAL TABLE

Flow: S3 (Data Lake) ❑————❑ External Table (Pointer)

Data STAYS here

Metadata ONLY

What happens:

- The data stays **only in S3** – it is **not moved anywhere**
- When we create an **External Table**, Snowflake creates **only a pointer** to the data
- **Only metadata** is stored in Snowflake – table structure and file locations
- When a query is executed, **Snowflake reads the data directly from S3**

```
CREATE OR REPLACE EXTERNAL TABLE ext_students (  
  student_id  NUMBER AS (VALUE:c1::NUMBER),  
  student_name STRING AS (VALUE:c2::STRING),  
  gender      STRING AS (VALUE:c3::STRING),  
  class       NUMBER AS (VALUE:c4::NUMBER),  
  section     STRING AS (VALUE:c5::STRING),  
  marks       NUMBER AS (VALUE:c6::NUMBER))  
  
WITH LOCATION = @stu_stg  
  
AUTO_REFRESH = TRUE  
  
FILE_FORMAT = students_csv_format;
```

Results (just now)	
Table	Chart
1 row	4.8s
1	Table EXT_STUDENTS successfully created.

2. What metadata does Snowflake store for external tables?

If data is in S3, what does Snowflake keep?

File list,schema,partitions(no data)

```
SELECT METADATA$FILENAME, METADATA$FILE_ROW_NUMBER
FROM ext_students LIMIT 5;
```

Results (2 minutes ago)

TableChart

5 rows

4.3s

	A METADATA\$FILENAME	# METADATA\$FILE_ROW_NUMBER
1	EXTERNAL FILES/students.csv	1
2	EXTERNAL FILES/students.csv	2
3	EXTERNAL FILES/students.csv	3
4	EXTERNAL FILES/students.csv	4
5	EXTERNAL FILES/students.csv	5

3. How does Snowflake read CSV files during query execution?

QUERY EXECUTION:

User Query — Metadata Check — Warehouse Start — S3 Request — Raw Data

Return — Parse Data — Apply Filter — Results

```
SELECT * FROM EXT_STUDENTS WHERE MARKS >=90;
```

Results (just now)

Table

Chart

2 rows

4.4s

#	{?} VALUE	# STUDENT_ID	STUDENT_NAME	GENDER	CLASS	SECTION	MARKS
1	{ "c1": "102", "c2": "Priya Sharma", "c3": "Female", "c4": "10"	102	Priya Sharma	Female	10	B	92
2	{ "c1": "105", "c2": "Karthik Rao", "c3": "Male", "c4": "10"	105	Karthik Rao	Male	10	A	90

Snowflake reads CSV files directly from S3 at query time using the virtual warehouse.

- Does querying an external table use warehouse compute?
- What happens if a file is deleted or modified in S3?
- How do you refresh external table metadata?
- Can external tables be used for transformations?
- How does schema-on-read differ from schema-on-write?
- What are the performance limitations of external tables?
- When should external tables be avoided?

● **Cross-Method Comparison (Critical Thinking)**

1. Which method provides the **highest security** and why?
2. Which method is best for **large historical data**?
3. Which method is best for **frequent small file ingestion**?
4. Which method gives the **best query performance**?
5. Which method has the **lowest operational overhead**?
6. How does cost vary between the four methods?
7. Which method aligns best with **data lake architecture**?
8. Which method would you choose for compliance-heavy environments?
9. How does failure handling differ across methods?
10. Which method would you recommend for enterprise production pipelines?